

FRONT WALL - SET UP LIST

- Put caulk by heater
- Caulking gun
- Framing nail gun
- Framing nails
- Panel nails
- Triangle
- Long Screws (brown)
- Short Screws (yellow)
- Screw gun with T25 bit
- Chalk line
- Stapler
- Staples
- Mallet
- Hammer

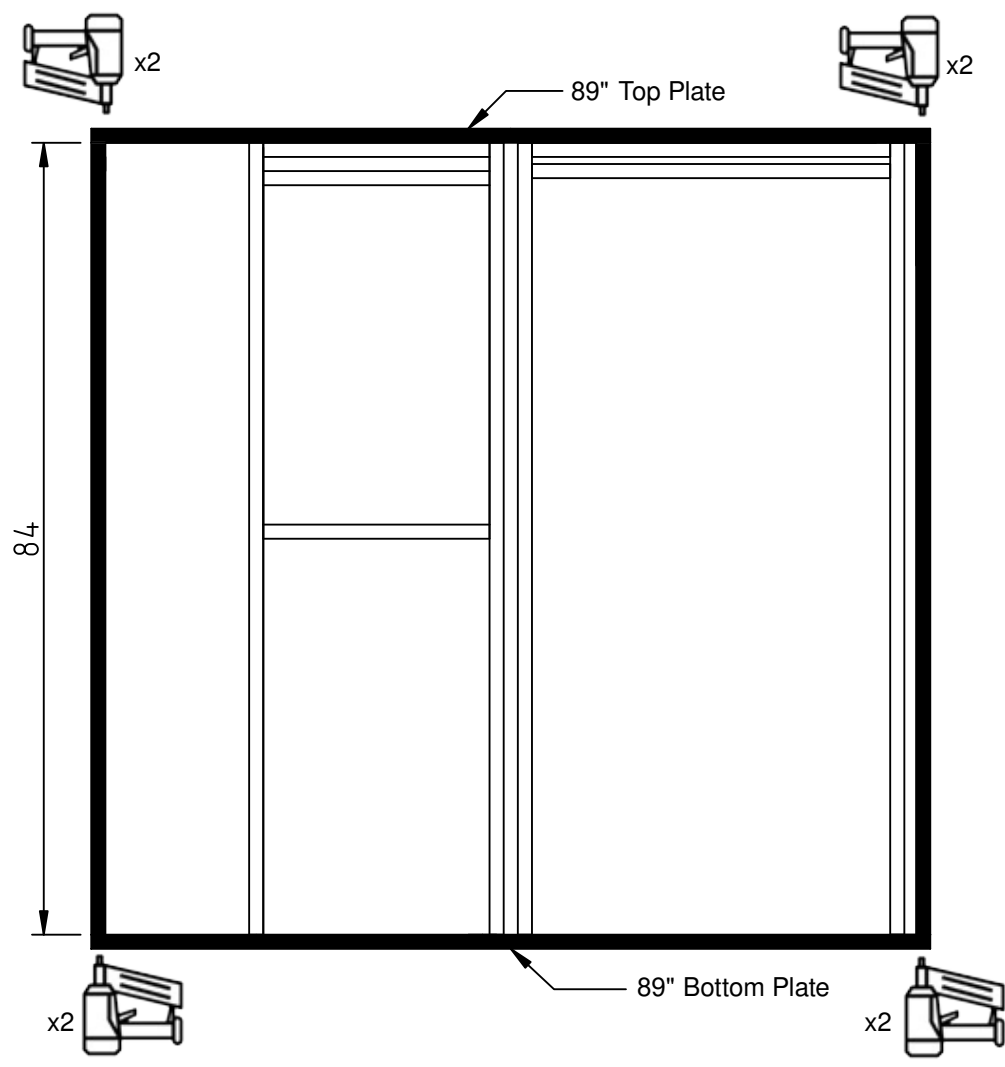
Step 1

- Place 2x4s in the designated spots on the perimeter.
- Line up the corners to be flush on top and sides
 - one person holds and the other nails.
- Put 1 FRAMING nail (the longer ones) in the upper middle of the stud to connect the corners.
- Check the perimeter squareness, by measuring the diagonal distance both ways. These should be equal. If not, adjust.
- Put a second nail directly under each of the first nails in the three nailed corners.

HAZARDS:

- Bashed fingers: Keep fingers 6" from mallet
- Nail Injury: Keep fingers 12" from joint

Top



Bottom

Step 1
Front Wall

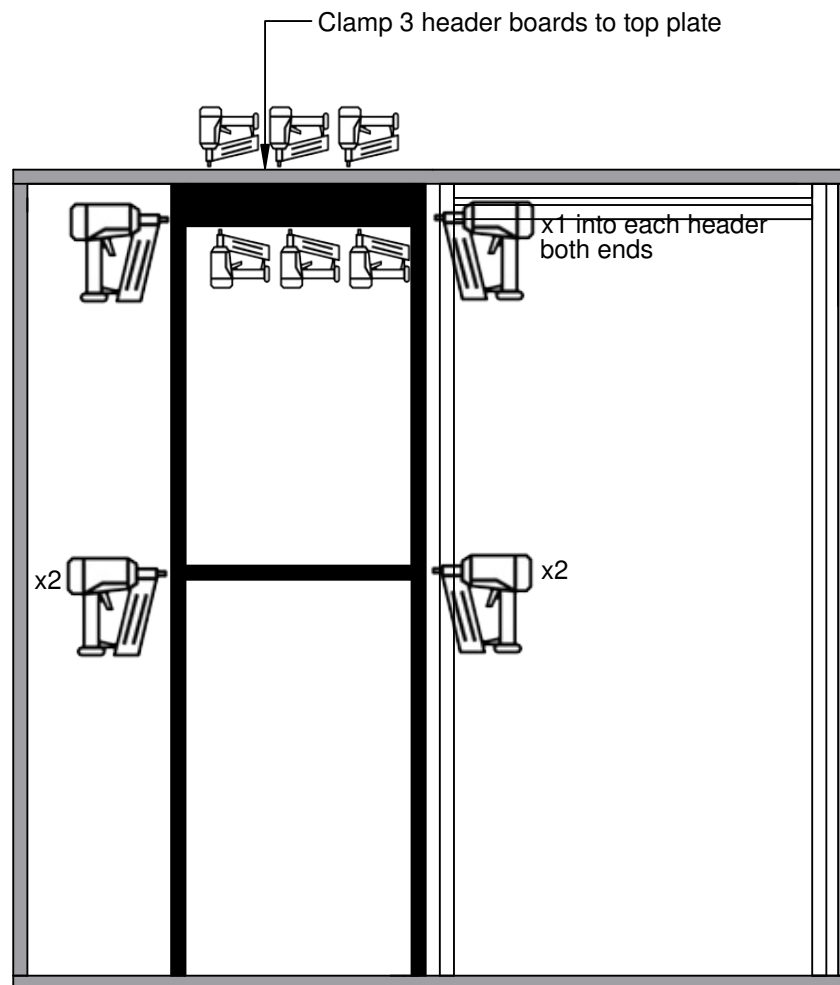
Step 2

- Place
 - inner 2 long 2x4s on both sides of the window area
 - 4 short 2x4s as shown
 - 3 short at the top and 1 in the middle.
- Clamp the 3 short boards at the top.
- Nail into both sides of the 3 short boards
 - 3 inside pointing out and 3 outside pointing in
- Squeeze the long boards (studs) up against the 3 short boards and put one nail through the stud on each side into the ends of each of the short boards.
- Put 2 nails into each end of the short 2x4 in the middle (through the studs at the bottom of the window).

HAZARDS:

- Bashed fingers: Keep fingers 6" from mallet
- Nail Injury: Keep fingers 12" from joint

Top



Bottom

Step 2
Front Wall

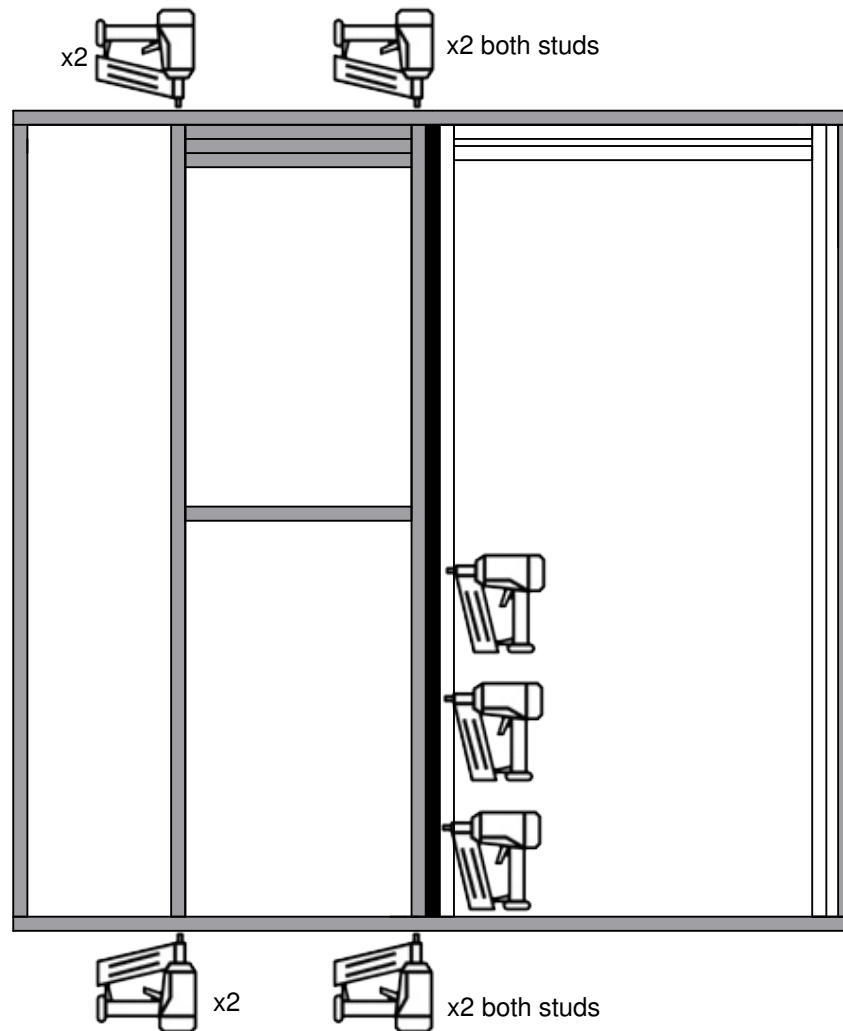
Step 3

- Team leader: When boards are in place, draw a straight vertical line at the middle of each stud to indicate where to put the nails
- Place the next stud as shown and clamp it to the adjacent stud. Nail these two studs together in three places below the window opening from the side opposite the window. If clamping does not make the joint tight, use 3" brown screws to force the two studs together below the window.
- Secure all 3 studs on both sides of the window
 - Use a clamp to squeeze any gap between the double studs
 - 2 nails through the top plate into each stud
 - 2 nails through the bottom plate into those same studs

HAZARDS:

- Bashed fingers: Keep fingers 6" from mallet
- Nail Injury: Keep fingers 12" from joint
- Ladder Fall: Ascend/descend ladder without carrying tools

Top



Bottom

Step 3
Front Wall

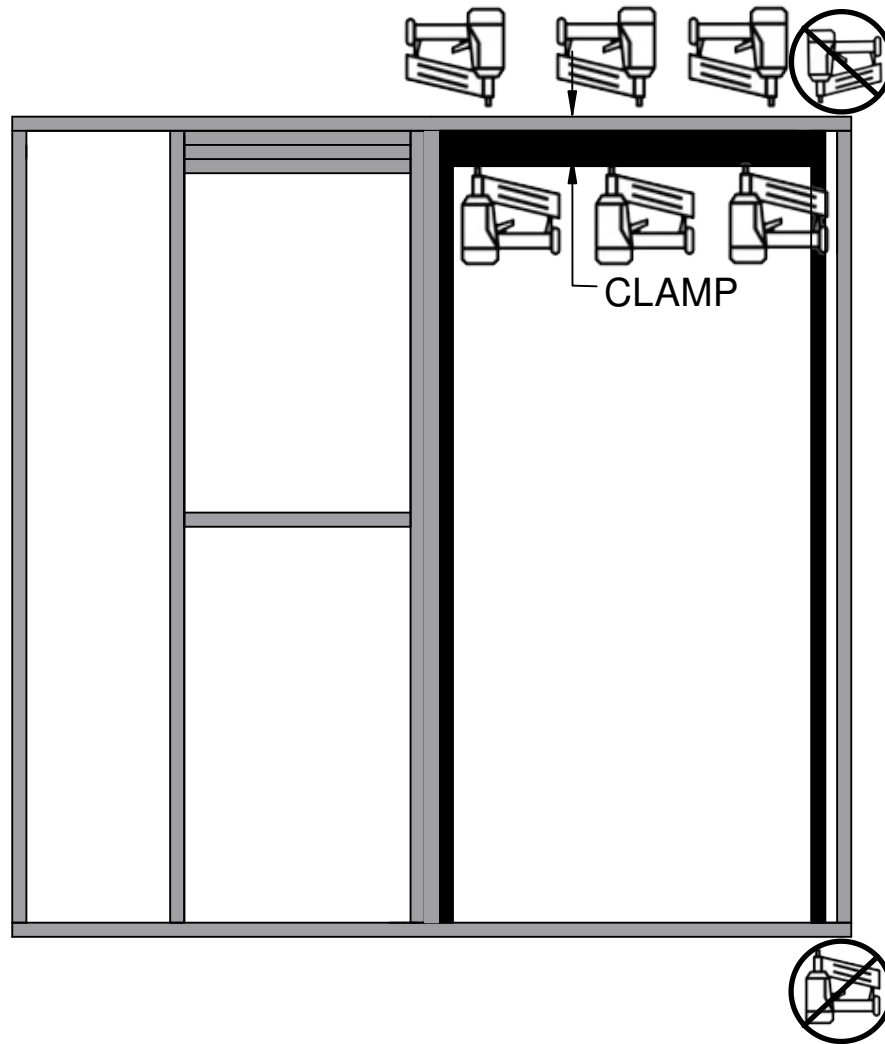
Step 4

- Place 2 long boards (studs), one on either side of the door opening, and two header blocks and a plywood spacer between them above the door opening.
- Push the long blocks above the door opening toward the window, and clamp them to the top plate. Nail as shown: 3 nails down through the top plate, and 3 nails up through the blocks.
- **Do not** nail into the ends of the studs on either side of the door opening. That will come in the next step.

HAZARDS:

- Nail Injury: Keep fingers 12" from gun end

Top



Bottom

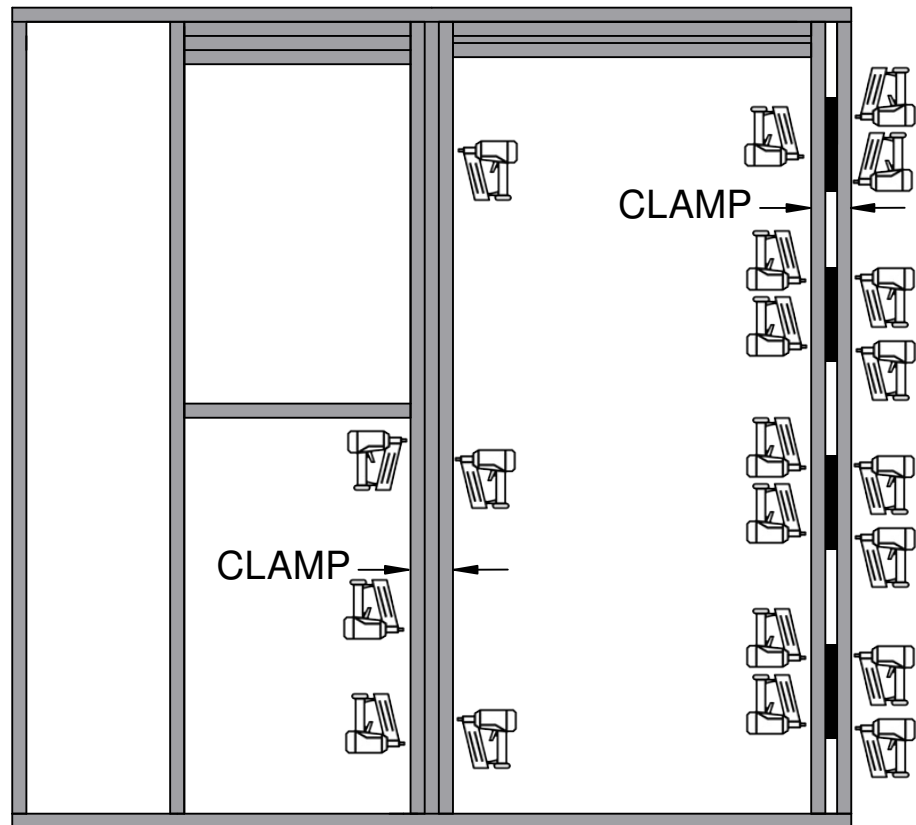
Step 4
Front Wall

Step 5

- Clamp the 3 studs in the center of the wall together, being certain that there is no gap, and nail as shown:
 - 3 nails from the window side, all below the window
 - 3 nails from the door side evenly spaced along the length of the stud. If there is any gap between the new stud and the two closer to the window, drive a 3" brown screw into the stud nearest the door and tighten it to remove that gap.
- Place 4 spacer blocks between the outer wall stud and the door-opening stud. Leave an 8" space above the top block and below the bottom block. Clamp each block between the two studs, and nail twice from each side.

HAZARDS: - Nail Injury: Keep fingers 12" from gun end

Top



Bottom

Step 5
Front Wall

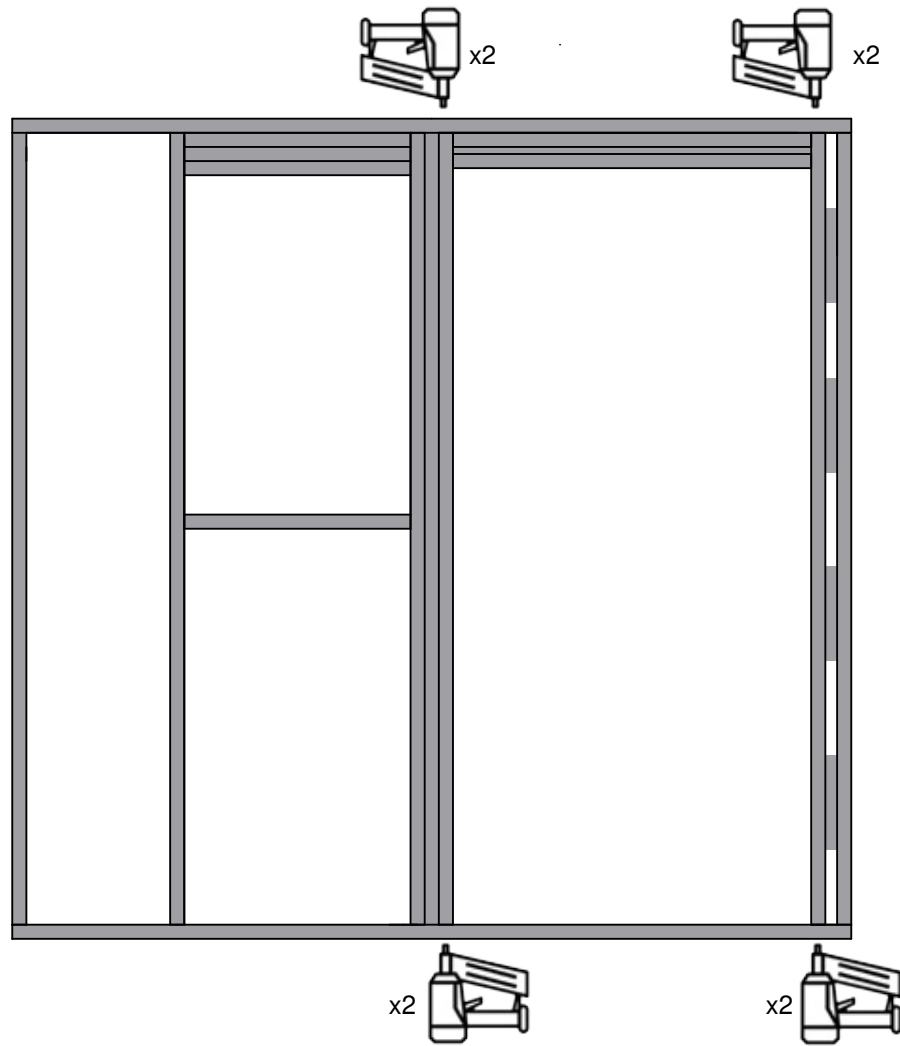
Step 6

- Put in the 2 nails through the top plate and 2 nails through the bottom plate into the studs on either side of the door opening.
- Check the perimeter to be sure all the long studs have 2 nails at each end.

HAZARDS:

- Nail Injury: Keep fingers 12" from gun end

Top



Bottom

Step 6
Front Wall

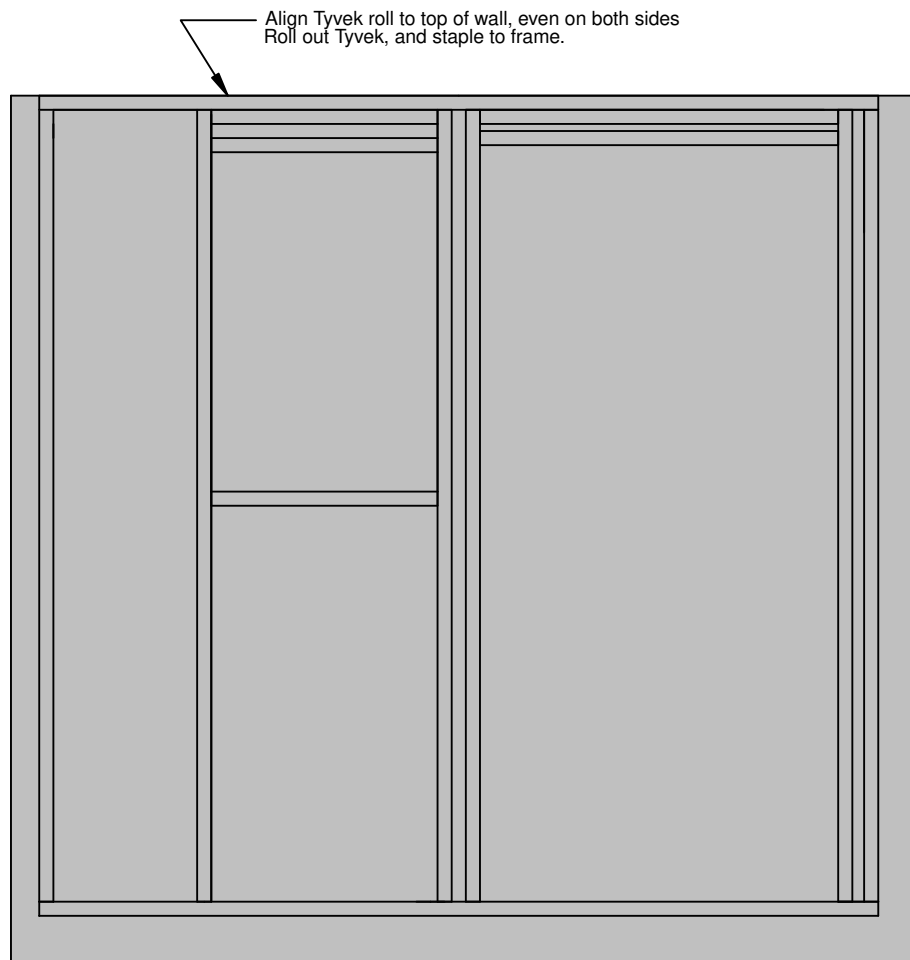
Step 7

- Line the Tyvek up to be flush with the top and overlap about 6" on the sides (outside the 2x4 guide posts).
- Roll out little bits at a time and staple 8" apart along the top stud, pulling it tight
— at the end of the top, cut off the roll with about 6 inches extra on the side.
- Staple around the perimeter going around in a circle pulling gently so that the Tyvek is tight.
- Staple along ALL inner studs, and across the top and bottom of the window opening and across the top of the door opening.

HAZARDS:

- Staple Injury: Keep fingers 12" from stapling point

Top



Bottom

Step 7
Front Wall

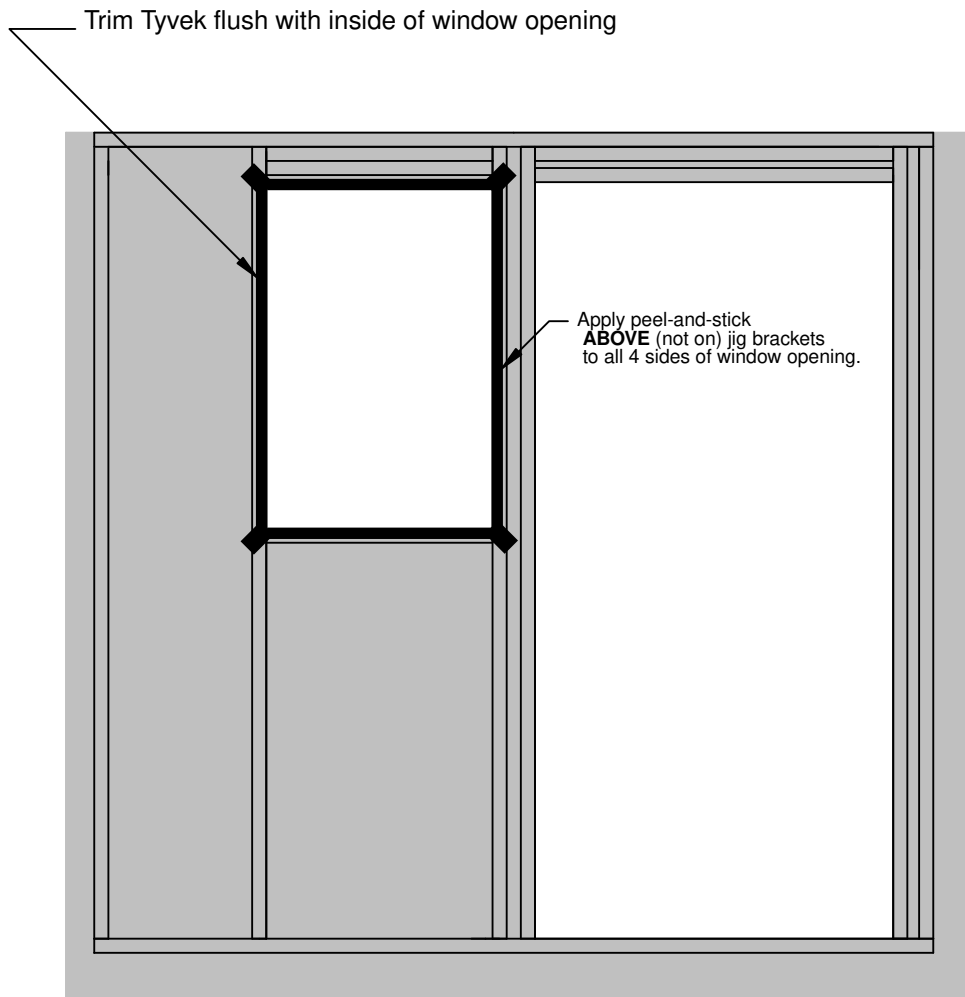
Step 8

- Cut the Tyvek out along the inside edge of the window.
- Install peel-and-stick flashing around the window opening
 - Using the pattern board, cut two “inner long” pieces and two “inner short” pieces
 - Apply one of the short lengths of flashing to the bottom of the window opening with about half of the flashing sticking up above the framing, wrapping the ends around the corners. Make sure that the flashing does not get stuck on the jig brackets
 - Be sure to press the flashing into each of the corners completely
- Use a knife to slit the flashing where it sticks up above the frame at each corner, and then fold the free section out over the frame and Tyvek.
- Apply a longer piece of flashing to each side of the window opening in the same way.
- Finally, apply the remaining short piece of flashing to the top of the window opening
- Cut 4 small pieces of the same flashing, and place one diagonally over each corner, as shown in the drawing. Force the end down into the corner of the window opening.
- **IMPORTANT** — Be sure to put the flashing ABOVE, not on the brackets of the jig.
- Cut the Tyvek out along the inside edge of the door opening. Save the cut-out piece; it can be used to create structural triangle covering later.

HAZARDS:

- Knife cut: Keep fingers clear of where the knife might slip

Top



Bottom

Step 8
Front Wall

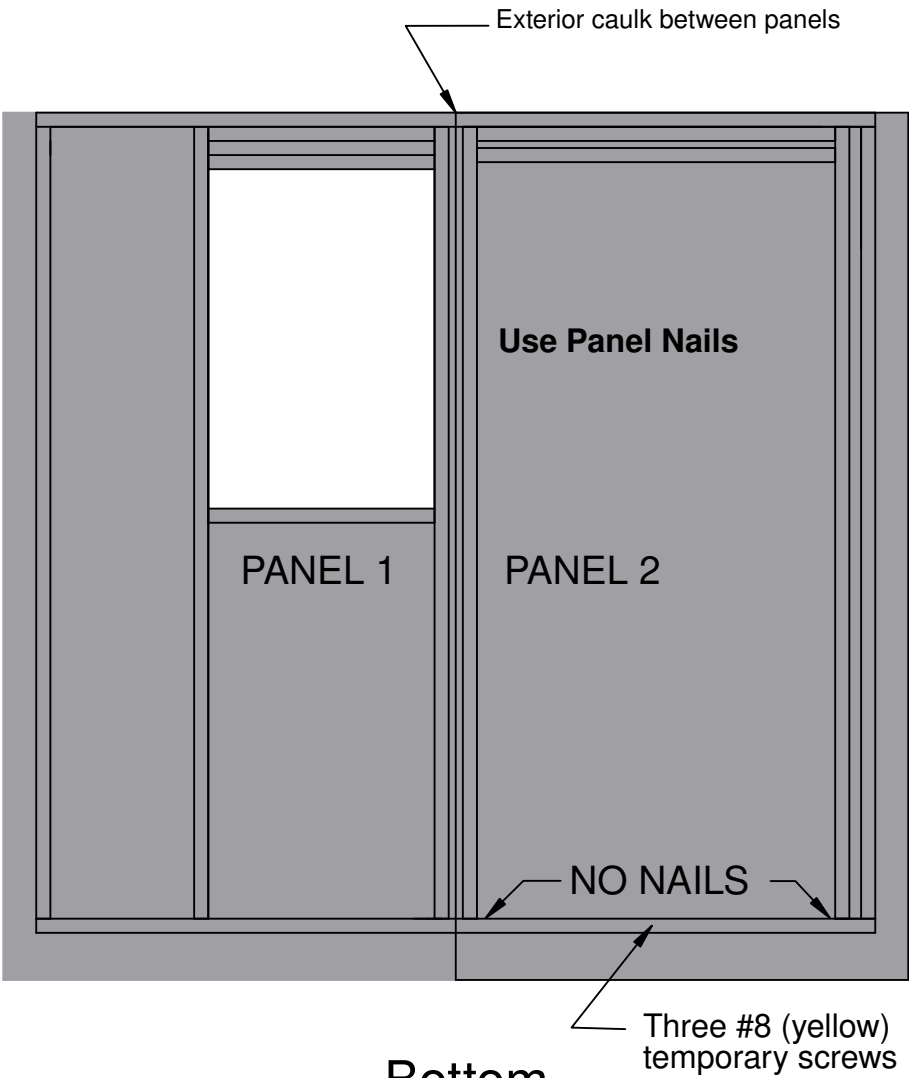
Step 9

- Place Panel 1 with window cut out. Align with top edge of frame and side stops.
- Nail (PANEL NAILS)
 - along top, sides and internal stud below the window.
 - use chalk lines
 - DO NOT nail around the window
- Caulk along the tongue edge of Panel 1.
- Place Panel 2, taking care that the joint between panels engages fully. Line panel up with top of frame.
- Team Leader: Use a ladder to climb onto the siding, and, using your body weight to press the seam down, nail every 8" along the joint. Keep nail heads ½" away from the seam to avoid splitting the siding. This probably means nailing at about a 30° angle, so that the nails anchor into the frame.
- Nail (PANEL NAILS)
 - along top and sides
 - use chalk lines
 - **DO NOT NAIL BELOW THE DOOR**
- Install three #8 (yellow) screws through the siding into the bottom plate. Be sure these do not stick up above the panel; they will interfere with the router. These screws will be removed after the wall is tilted up.

HAZARDS:

- Nail Injury: Keep fingers 12" from joint
- Ladder Fall: Ascend/descend ladder without carrying tools

Top



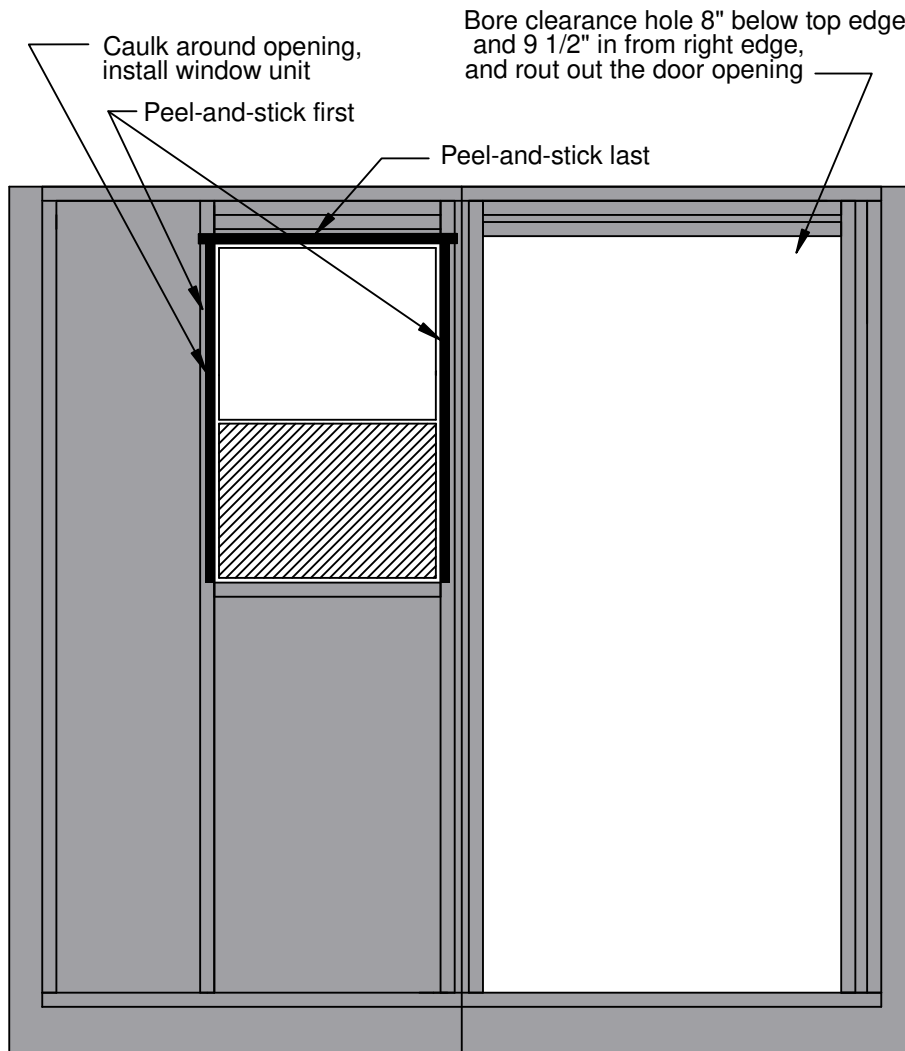
Bottom

Step 9
Front Wall

Step 10

- Team leader: Bore a hole through the siding where the arrows on the jig indicate, and rout out the door opening (do this now, to avoid fouling the router plate with caulk later.)
- Prepare the window. Remove all the labeling, and fully open the window. Set it screen-side down on the paneling.
- Apply caulk down the middle of the window flanges on the top and both sides
 - NOT at the bottom.
 - caulk on the opposite side from the screen (the surface that will lie on the panel)
- Place the window in the opening
 - SCREEN at the BOTTOM facing up
- Put SHORT SCREWS around the window.
 - 4 on each side and 3 on the top and bottom.
- Team leader: Using a piece of 1x4 trim, draw a pencil line on all four sides of the window to show where the trim will go.
- Put peel-and-stick around 3 sides of the window
 - **NOT** on the bottom of the window
 - Using the pattern board, cut two pieces “outer long” and one piece “outer short”
 - Place all peel-and-stick FLUSH against the edge of the window
 - Make sure the peel-and-stick does not extend over the pencil outline

Top



Bottom

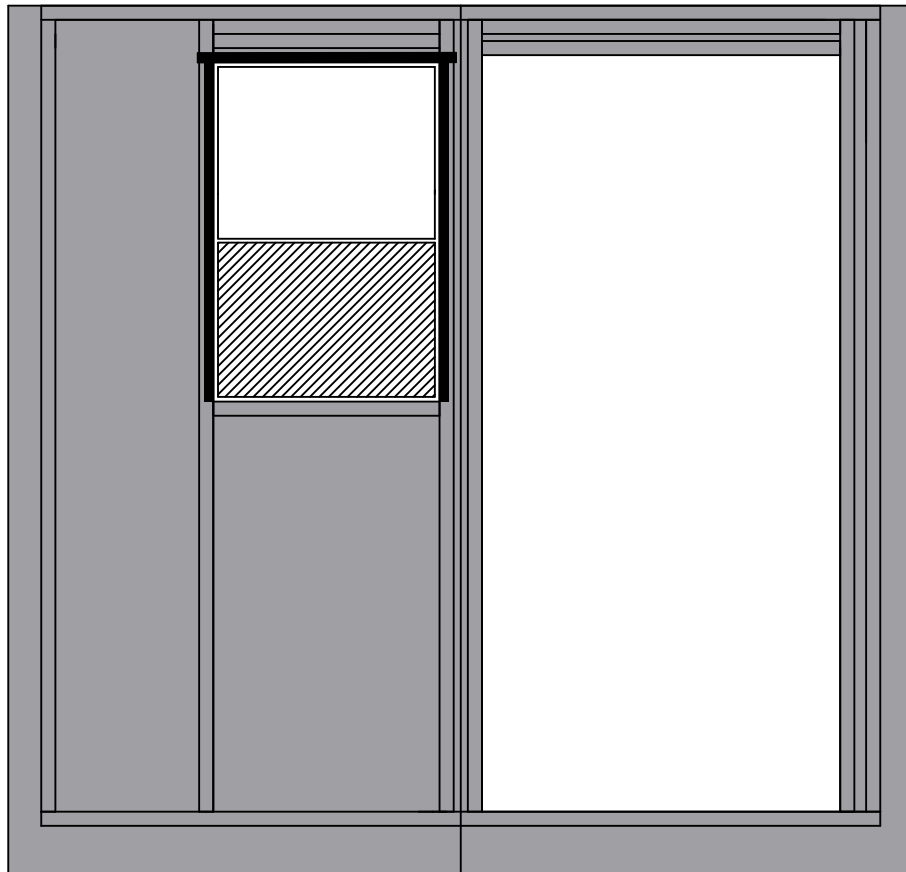
Step 10
Front Wall

Step 11

- Pry gently on the top and bottom plates to raise the wall out of the jig, and place blocks under these boards.
- Staple Tyvek to underside of panels on the bottom and both sides. Trim Tyvek on all 3 sides.
- Nail every 8" on the two sides and top of the door opening.
- Caulk all nails after tilt-up.

Top

Nail sides and top of door opening,
every 8". DO NOT nail across
bottom of door opening.



Bottom

Step 11
Front Wall

FRONT WALL CHECKLIST

1. Perimeter 2x4s — nail corners
2. 2x4 studs next to window
3. 3 short 2x4s above window — CLAMP, nail
4. 1 short 2x4 below window, nail
5. 2 nails at both ends of long 2x4s next to window
6. Inner studs
7. CLAMP double studs - 2 nails **BELOW** window ONLY
8. Small 2X4s over door, nail
9. 3rd 2x4 stud in center of wall; nail
10. 4 small blocks at right side of door using CLAMP
11. 2 nails at the ends of all long studs
12. Tyvek staple perimeter and inside studs
13. Cut out Tyvek for window and door
14. Peel & Stick edge of window (**NOT** on jig brackets)
15. Panel 1 with window hole, nail (PANEL NAILS)
16. Caulk edge of panel 1
17. Panel 2 nail top, sides and internal studs **NOT below door**
18. Caulk edge of window, and place window into opening
19. Short SCREWS — 4 on sides, 3 top and bottom
20. Rout door opening
21. Peel & stick 3 edges of window, **NOT** under window
22. Staple and trim Tyvek
23. Nail panel along sides and top of door opening
24. Caulk nails after tilt-up